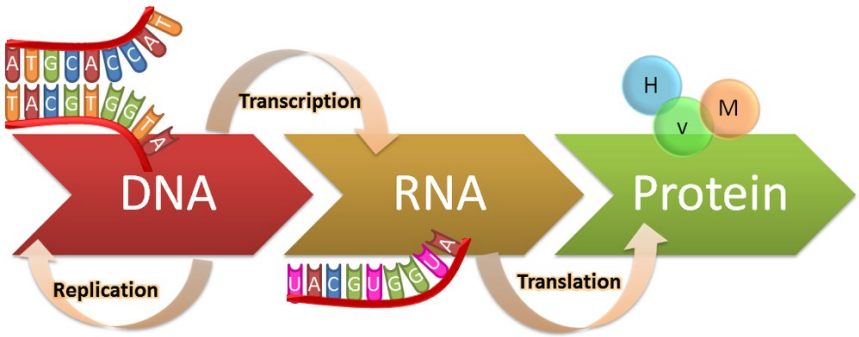
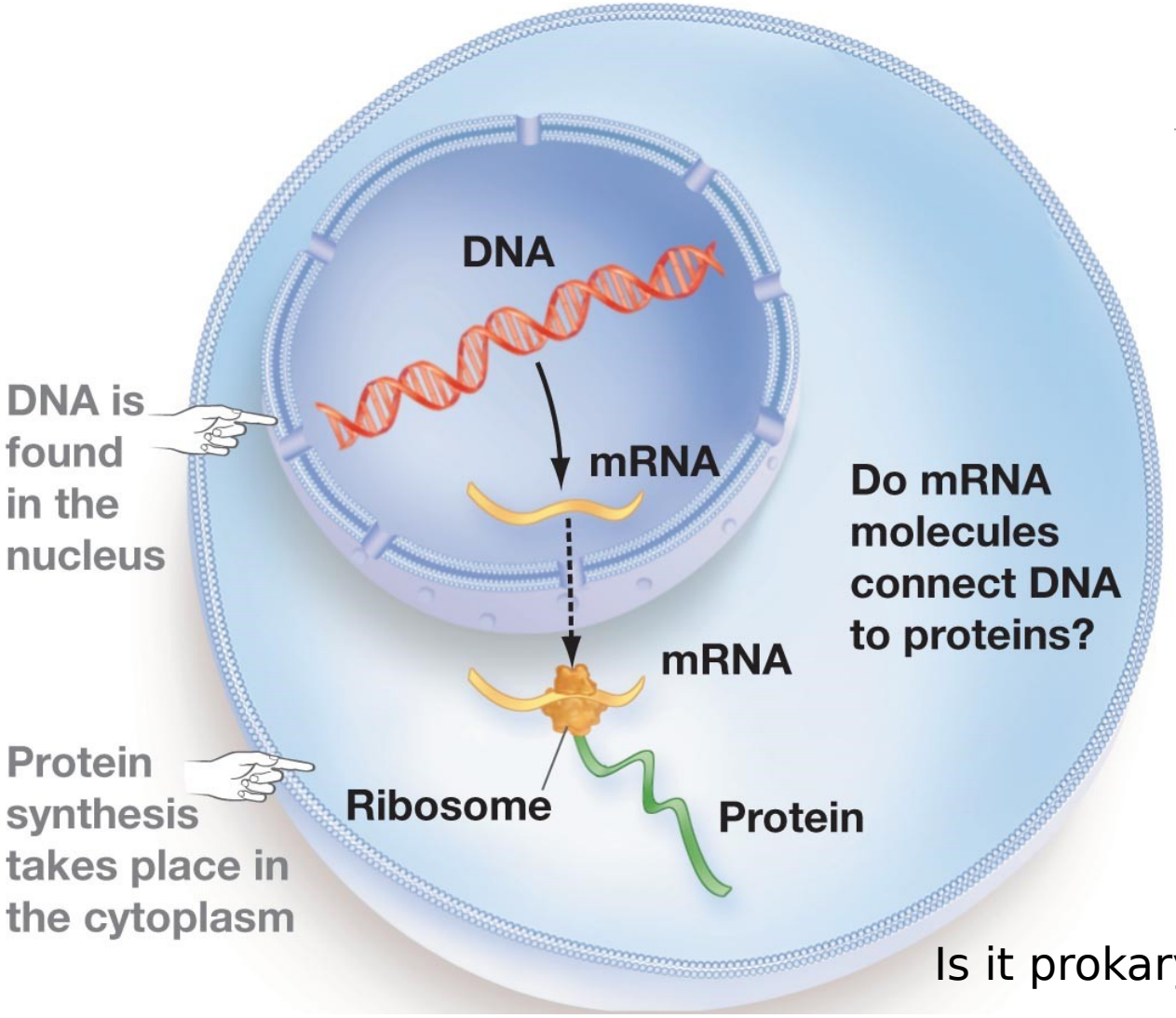


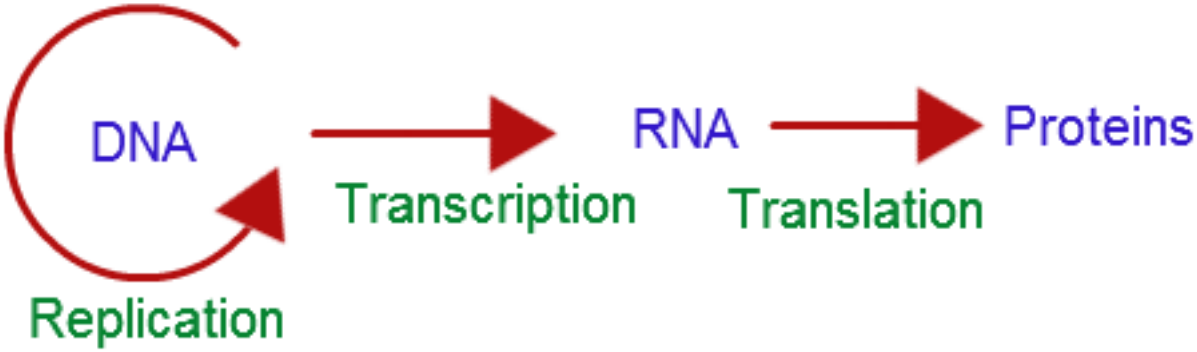
Central Dogma & DNA Replication

Central Dogma



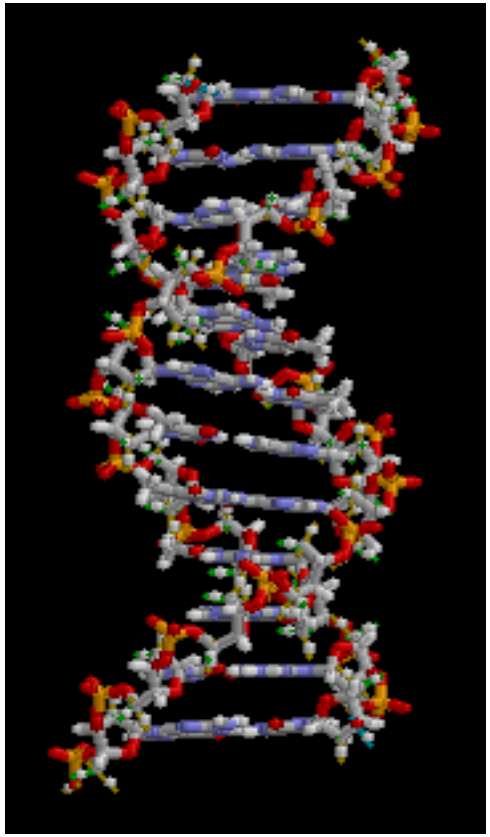
Is it prokaryotic or eukaryotic ce

Central Dogma- The Flow of Genetic Information



Transcription refers to RNA synthesis using DNA as a template

Translation is the synthesis of protein using the information in the RNA

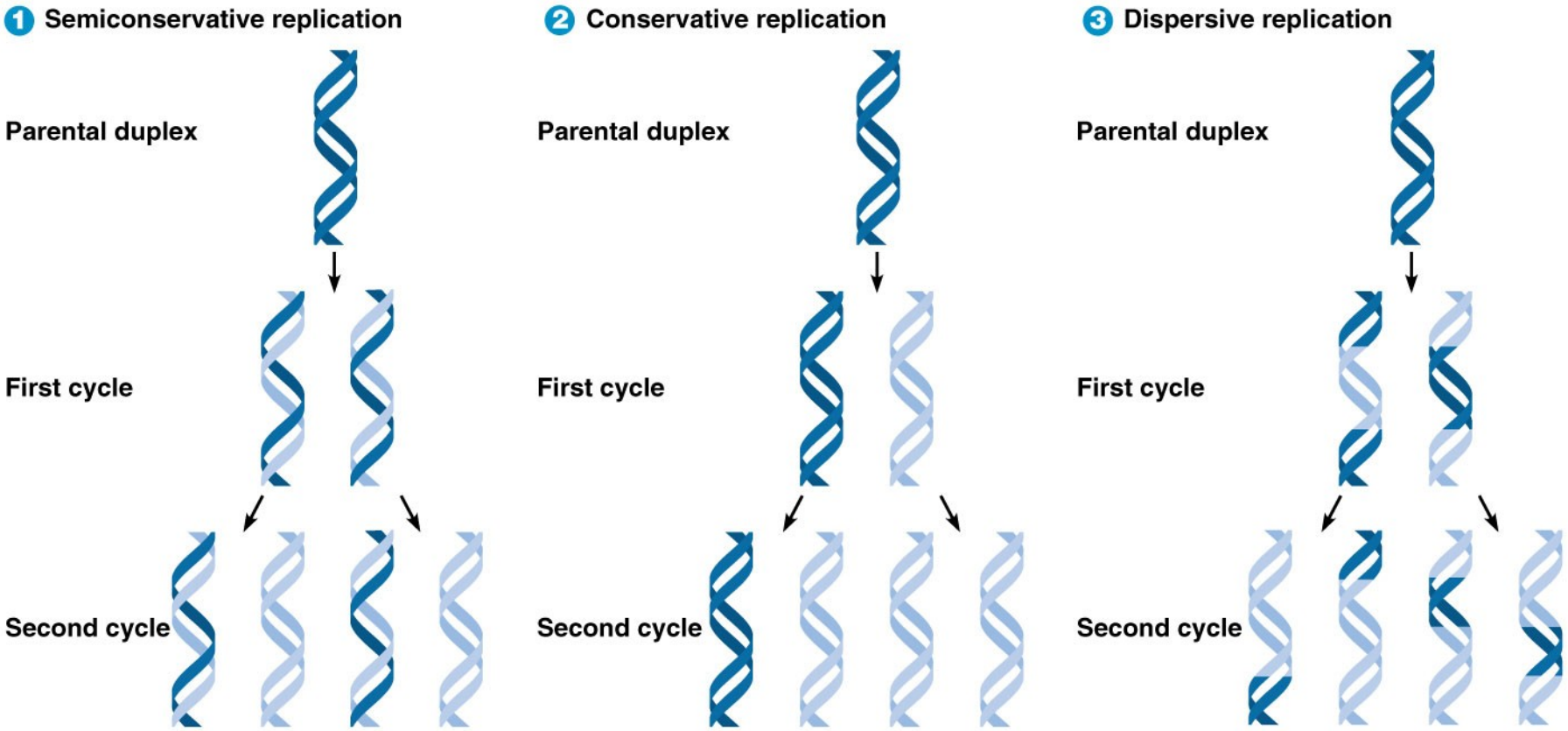


DNA Replication

DNA Replication, Repair, and Recombination

- Cells must be able to accurately reproduce, or *replicate,* their genetic material at each cell division
  - Hereditary nature of the genetic material
- Without accurate replication, the genetic material of resulting cells would be filled with errors
- Cells must also *repair* damage to their genetic material

How is the genetic material transmitted?



Demonstration of Semiconservative Replication

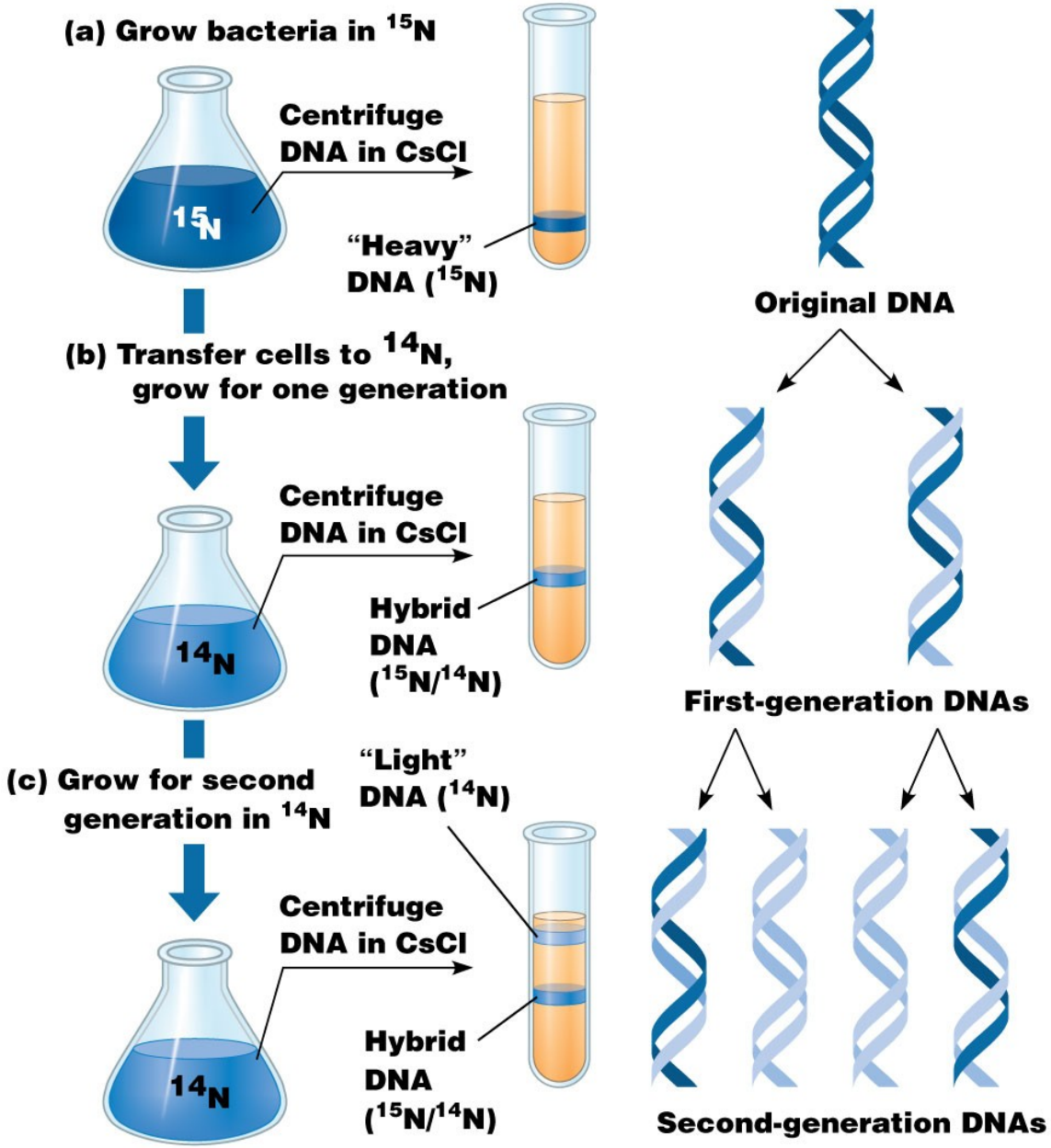
- Matthew Meselson and Franklin Stahl (with Jerome Vinograd) showed that replication is semiconservative by using **^{14}N and ^{15}N** to distinguish newly formed DNA strands from old
- Bacterial cells were grown in ^{15}N medium for many generations to incorporate heavy nitrogen into their DNA, then transferred to ^{14}N medium
- The strands were distinguished by *equilibrium density centrifugation*

Results of the Experiment

- After one cycle of replication in the ^{14}N medium, a band was observed intermediate between the heavy and light strands

- If the hybrid DNA was heated to separate the strands, one strand was “heavy,” and the other was “light”

- These results are consistent with **semiconservative replication**


(a) Grow bacteria in ^{15}N
Centrifuge DNA in CsCl
^{15}N
"Heavy" DNA (^{15}N)
Original DNA
(b) Transfer cells to ^{14}N, grow for one generation
Centrifuge DNA in CsCl
^{14}N
Hybrid DNA ($^{15}N/^{14}N$)
First-generation DNAs
(c) Grow for second generation in ^{14}N
"Light" DNA (^{14}N)
Centrifuge DNA in CsCl
^{14}N
Hybrid DNA ($^{15}N/^{14}N$)
Second-generation DNAs

Postulated mechanism of DNA replication

Replication of DNA as suggested by Watson and Crick.

The preexisting or “parent” strands become separated, and each is the template for biosynthesis of a complementary “daughter” strand

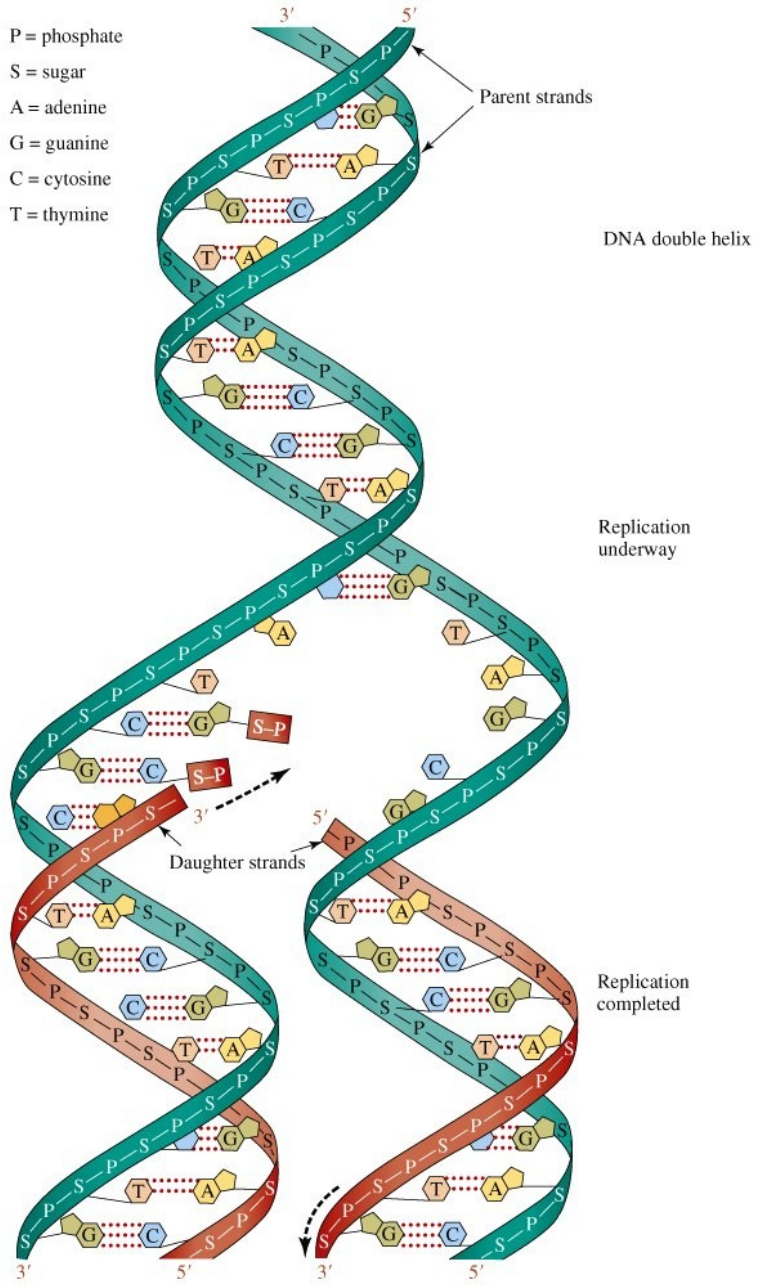
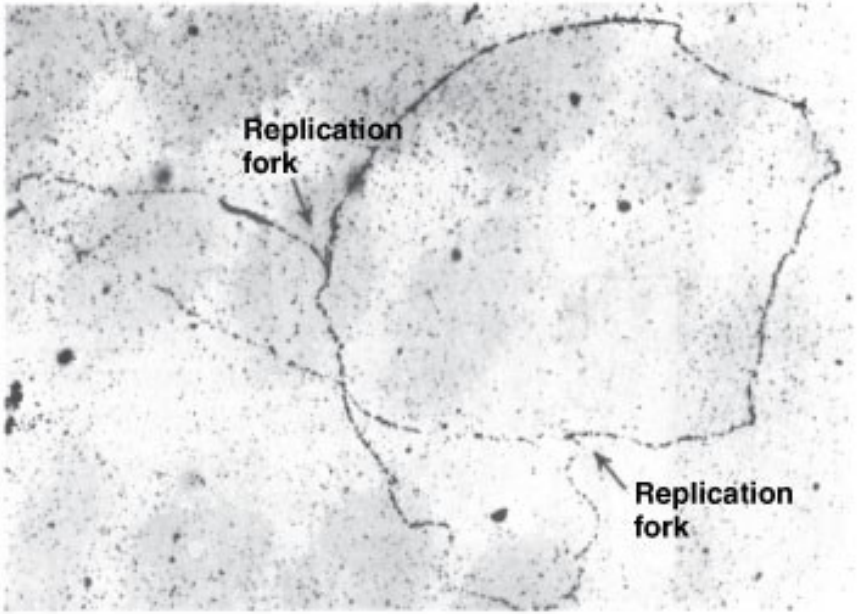


Figure 10-11 Concepts in Biochemistry, 3/e
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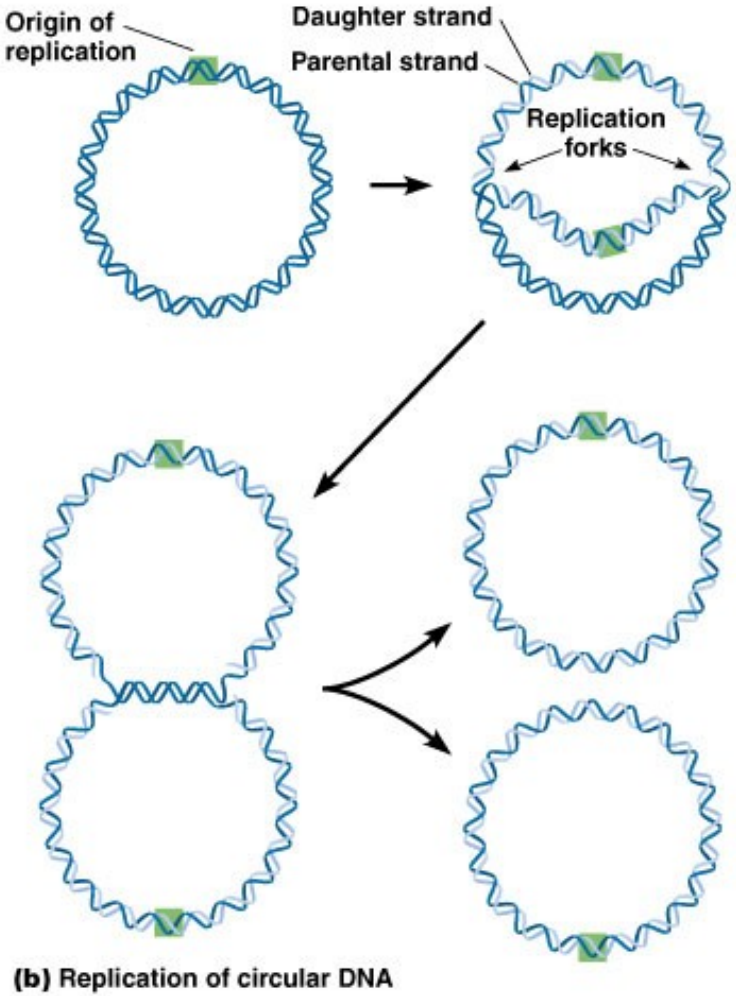
Prokaryotic Replication



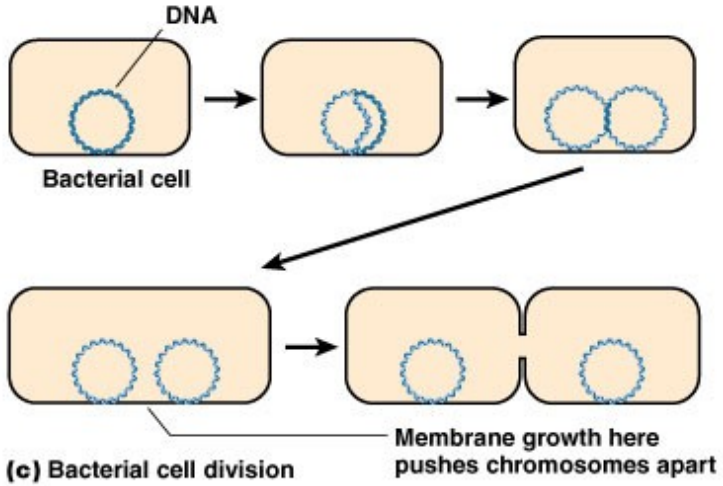
(a) Autoradiograph of *E. coli* DNA replication

Replication of a circular DNA molecule begins at a single origin and proceeds bidirectionally around the circle, with the two replication forks moving in opposite directions.

Prokaryotic Replication



(b) Replication of circular DNA

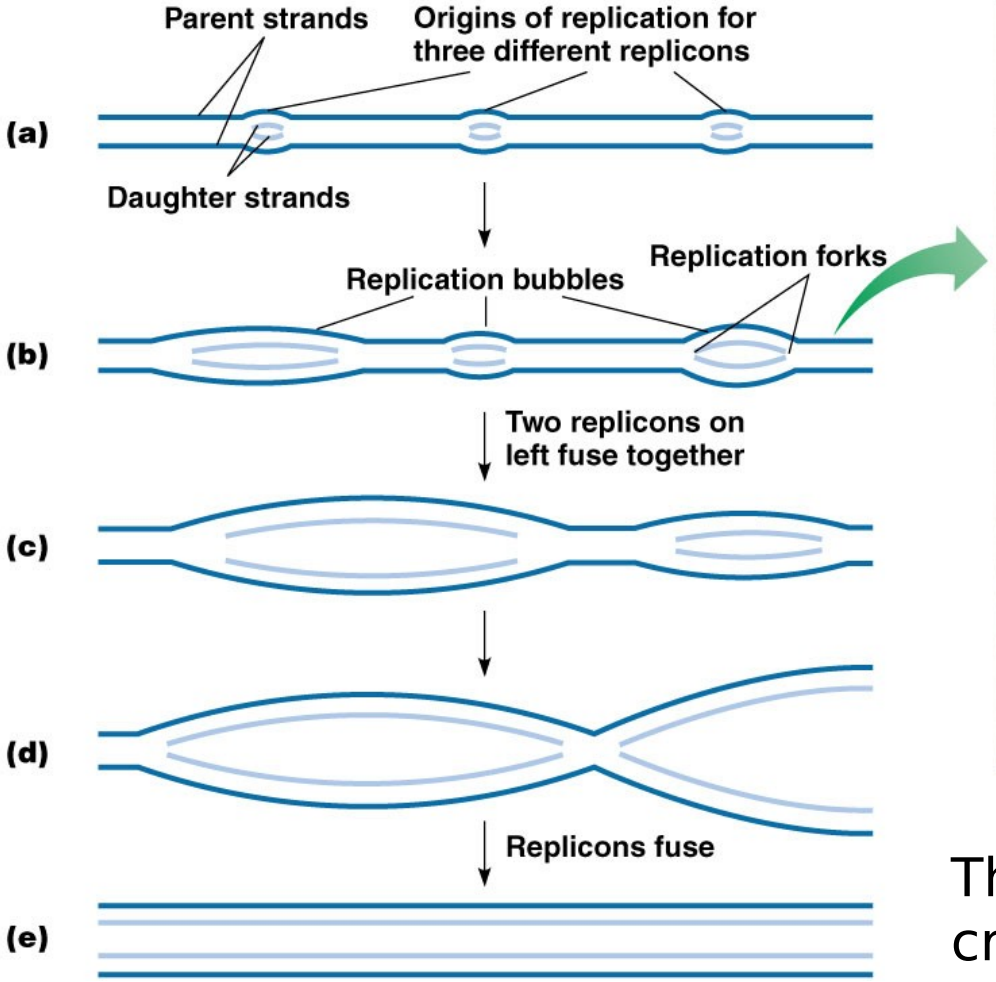


(c) Bacterial cell division

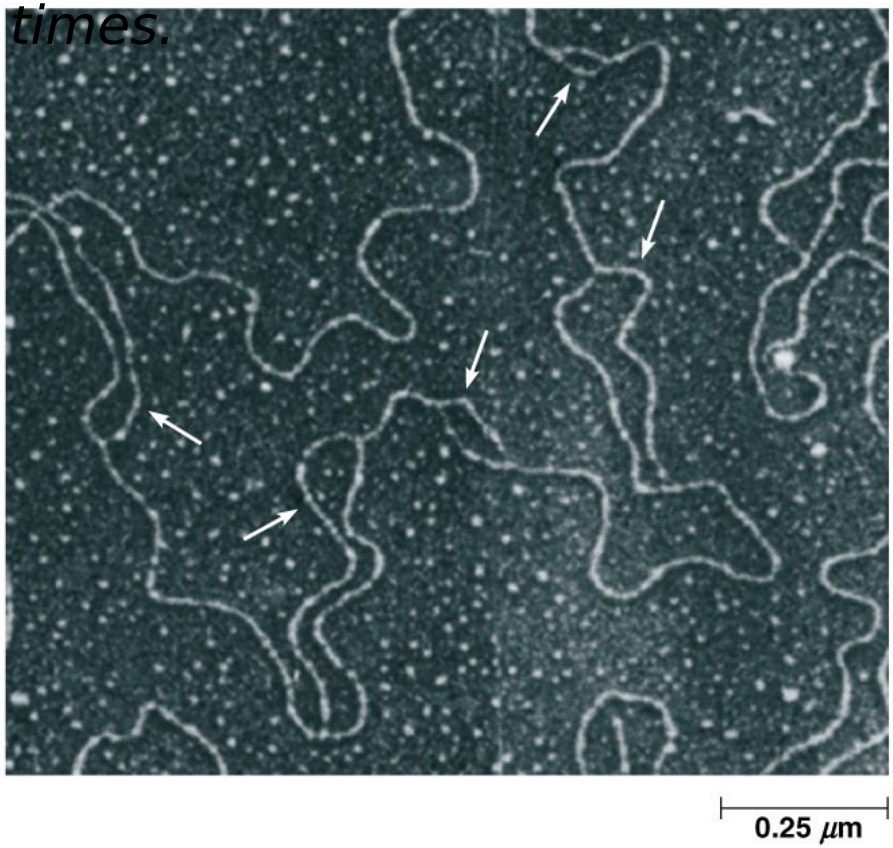
The new strands are shown in light blue.

Eukaryotic DNA Replication

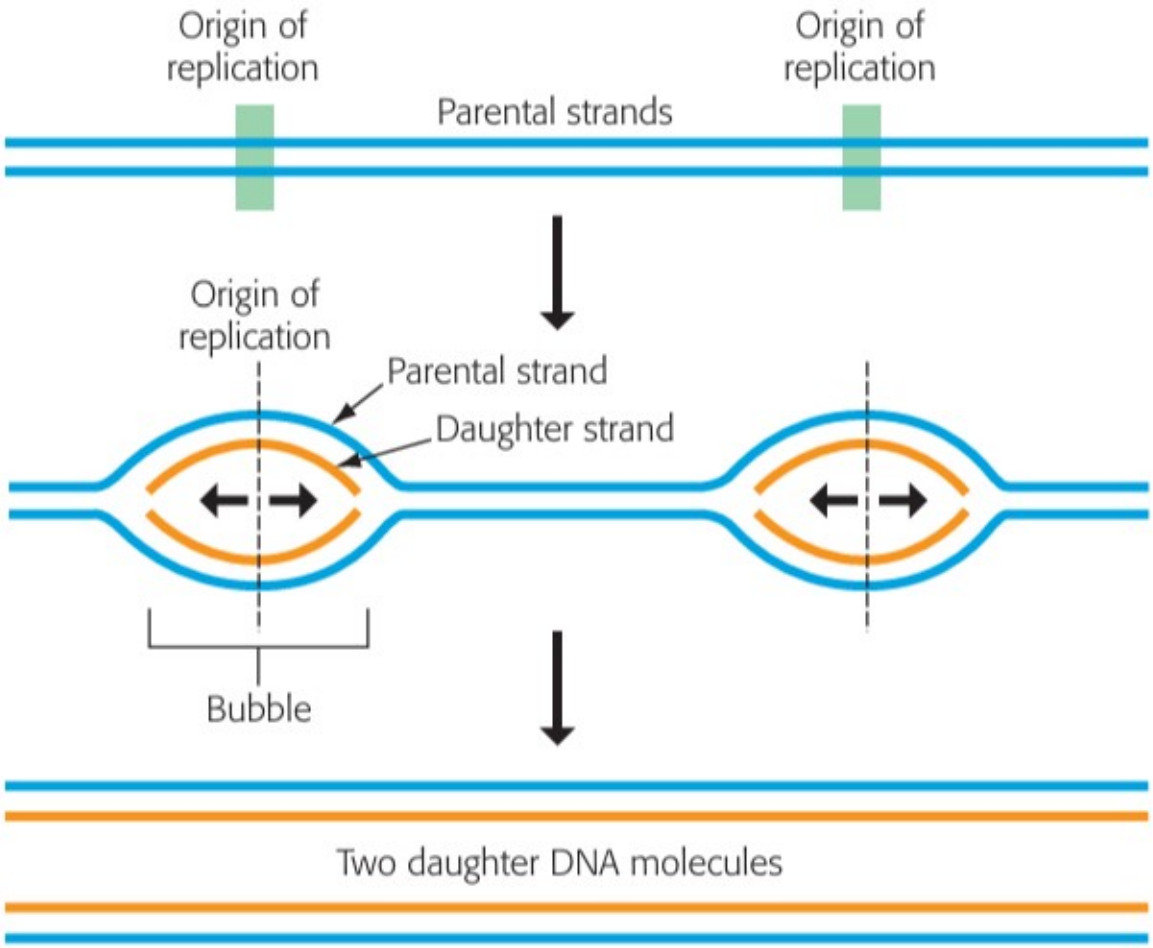
- In eukaryotes, replication of linear chromosomes is initiated at multiple sites, creating replication units called **replicons**
- The DNA of a typical chromosome may contain several thousand replicons, each 50,000 to 300,000 bp in length
- At each origin of replication, two replication forks synthesize DNA in opposite directions, forming a "*replication bubble*"

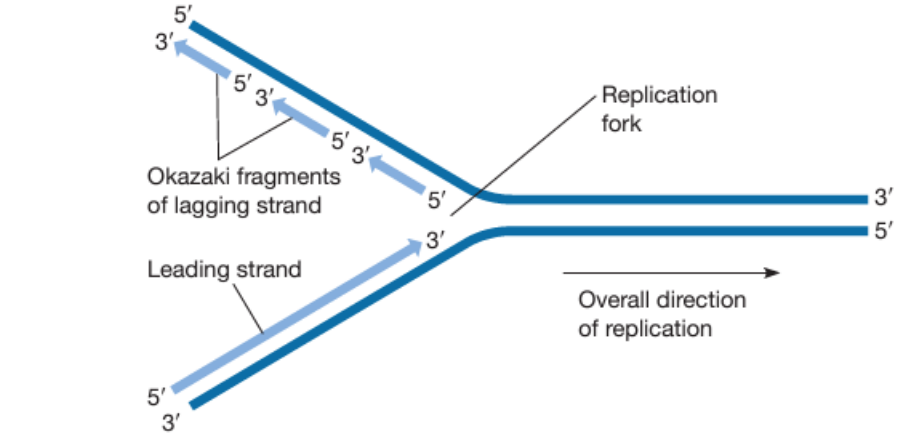


Origins can "fire" at different times.

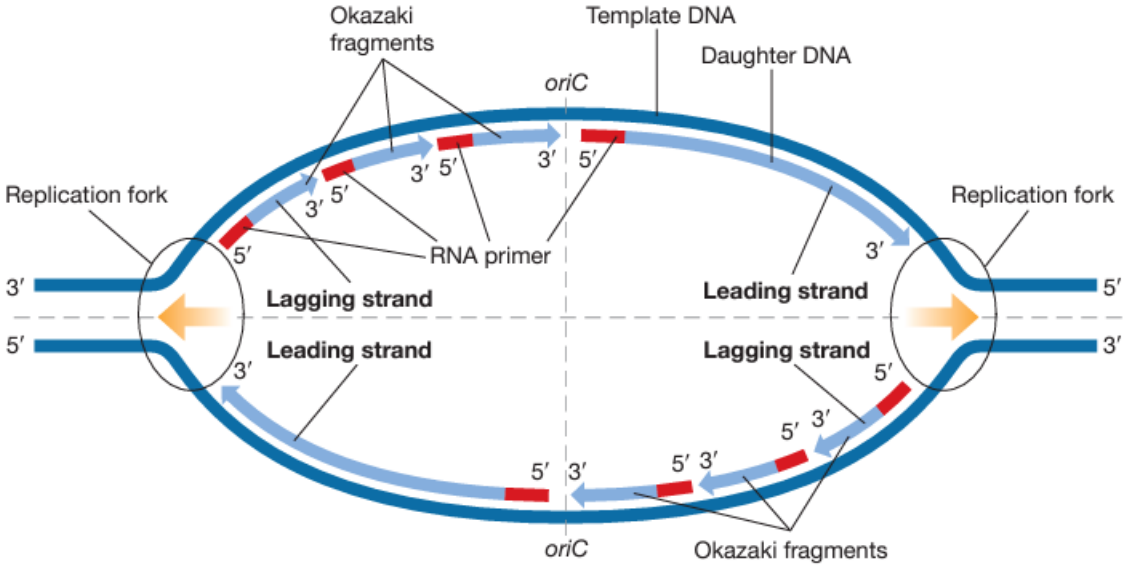


These replication forks are created by a DNA replication mechanism that begins at a single point within the DNA and proceeds in a bidirectional fashion away from this origin.


Origin of replication
Origin of replication
Parental strands
Origin of replication
Parental strand
Daughter strand
Bubble
Two daughter DNA molecules



(a) Replication fork



(b) Replication bubble

Figure 17-11 Directions of DNA Synthesis at a Replication Fork. (a) Replication fork structure. Parental DNA is shown in dark blue and newly synthesized DNA in lighter blue. Here and in subsequent figures, arrows indicate the direction in which the nucleic acid chain is being elongated. Because DNA polymerases synthesize DNA chains only in the 5′ → 3′ direction, synthesis at each replication fork is continuous in the direction of fork movement for the leading strand but discontinuous in the opposite direction for the lagging strand. Discontinuous synthesis involves short intermediates called Okazaki fragments, which are later joined together by the enzyme DNA ligase. **(b)** Relation of replication forks to replication bubble progression. Bidirectional expansion is driven by DNA synthesis at each replication fork.

Replication Initiates at Specialized DNA Elements

- The site where DNA replication initiates is known as an **origin of replication,** where synthesis is initiated by several groups of *initiator proteins*
- This sequence is AT rich and about 245 bp in length
- The sequence varies among bacterial species but contains recognizable, similar sequences, called **consensus sequences**

Initiation factors

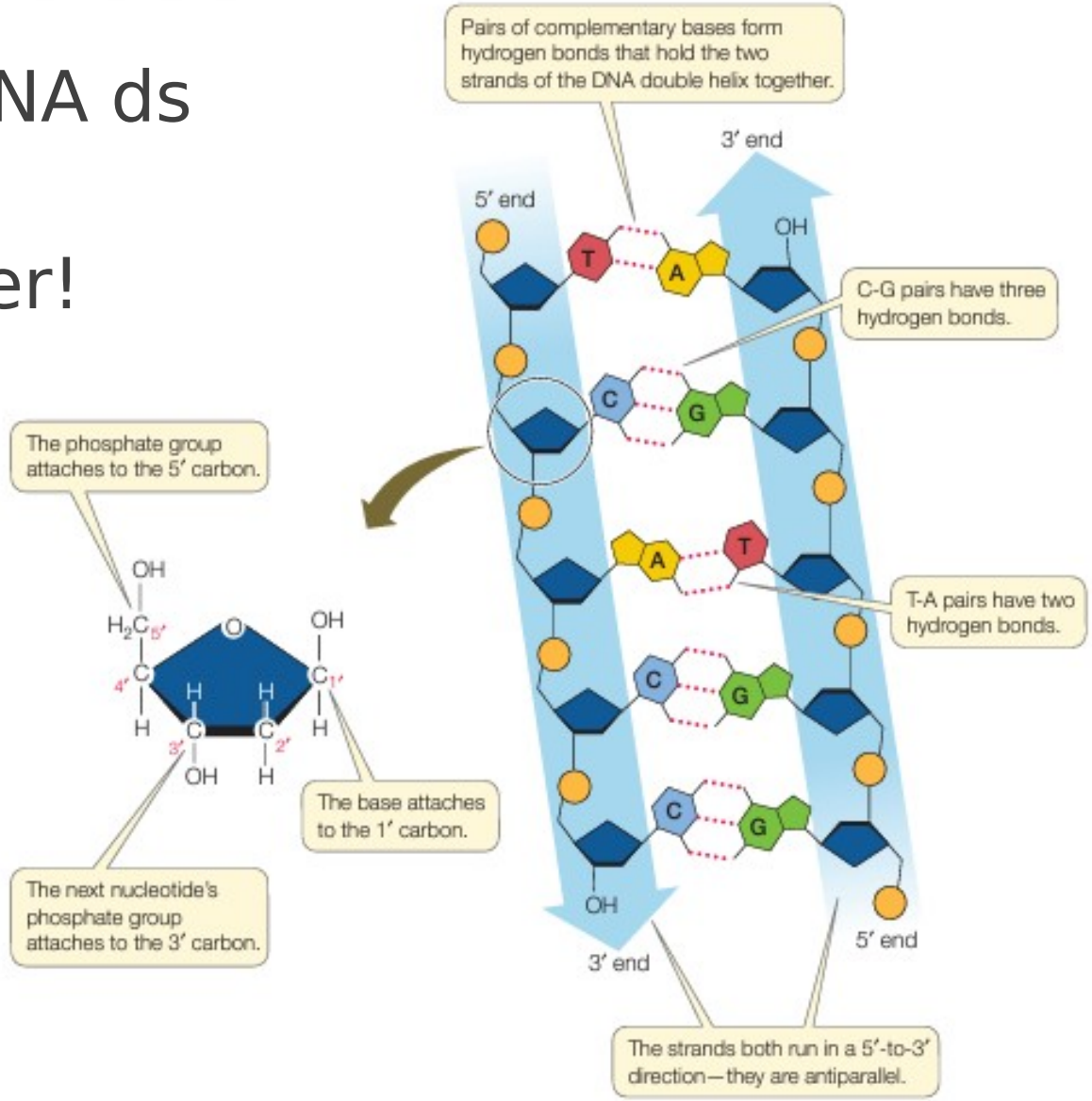
- Three functions:
  - Recognition of origin sequences (Origin recognition complexes)
  - DNA unwinding (helicase)
  - DNA synthesis and proofreading (DNA Polymerase)

  - Single stranded DNA is stabilized by special proteins during replication

DNA Polymerases Catalyze the Elongation of DNA Chains

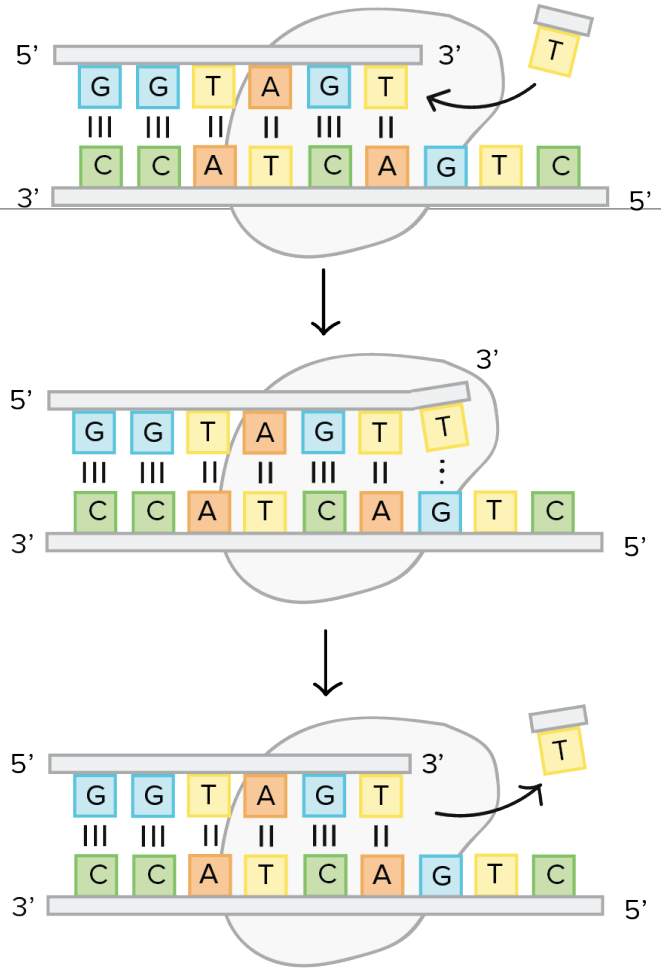
- **DNA polymerase** is an enzyme that can copy/synthesize DNA molecules
- Incoming nucleotides are added to the 3′ hydroxyl end of the growing DNA chain, so *elongation occurs in the 5′ to 3′ direction.*
- Several other forms of DNA polymerase have been identified; the original is now called *DNA polymerase I*

- Structure of DNA ds helix
- Let's remember!



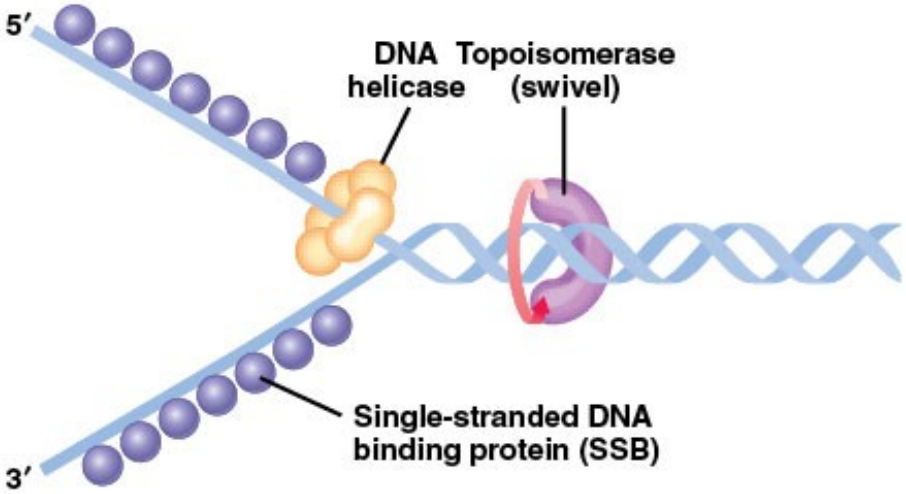
In Bacteria, Proofreading is Performed by the 3′→ 5′ Exonuclease Activity of DNA Polymerase

- About 1 of every 100,000 nucleotides incorporated during DNA replication is incorrect.
- Such mistakes are usually fixed by a **proofreading** mechanism
- Almost all DNA polymerases have a **3′ → 5′ exonuclease activity**

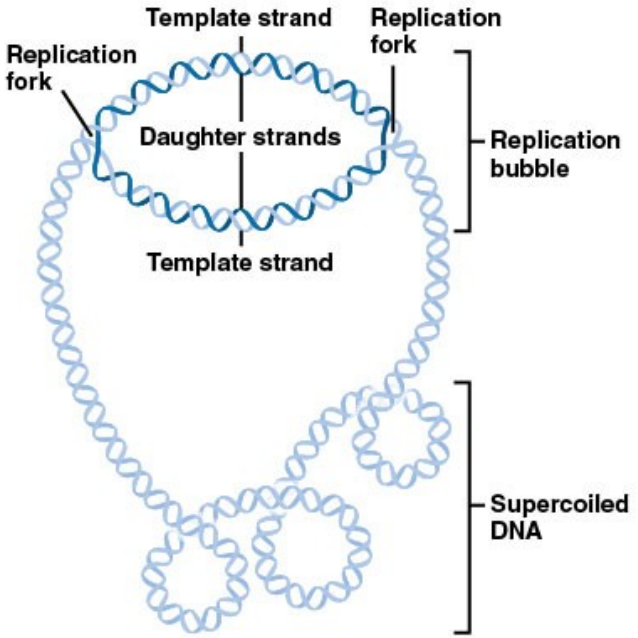

5’
3’
G G T A G T
C C A T C A G T C
3’
5’
T
Polymerase adds an incorrect nucleotide to the new strand of DNA.
5’
3’
G G T A G T T
C C A T C A G T C
3’
5’
Polymerase detects that bases are mispaired.
5’
3’
G G T A G T
C C A T C A G T C
3’
5’
T
Polymerase uses 3 ’→ 5’ exonuclease activity to remove incorrect nucleotide.

The DNA Double Helix Must Be Locally Unwound During Replication

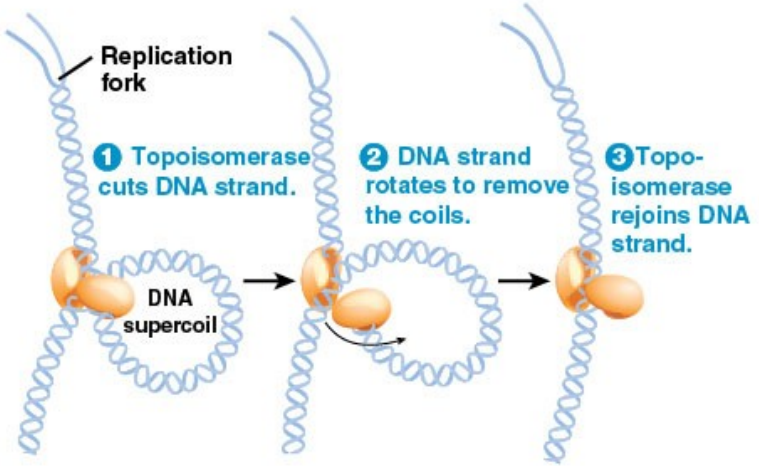
- During DNA replication, the two strands of the double helix must unwind at each replication fork
- Three classes of proteins facilitate the unwinding: *DNA helicases, topoisomerases,* and *single-stranded DNA binding proteins*
- **DNA helicases** are responsible for unwinding the DNA, using energy from ATP hydrolysis


5′
DNA helicase
Topoisomerase (swivel)
Single-stranded DNA binding protein (SSB)
3′

(a)


Template strand
Replication fork
Replication fork
Daughter strands
Replication bubble
Template strand
Supercoiled DNA

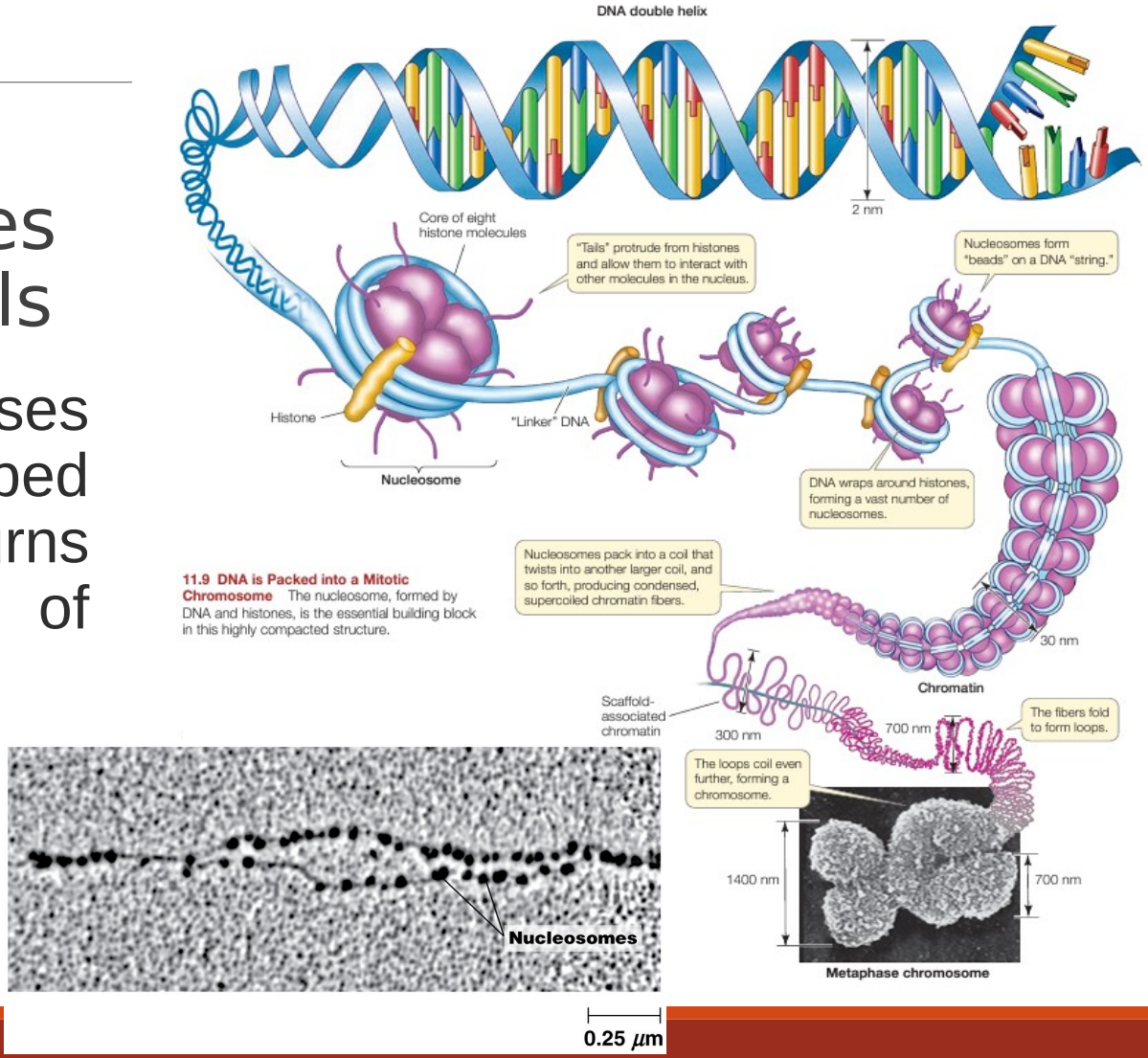
(b)


Replication fork
1 Topoisomerase cuts DNA strand.
2 DNA strand rotates to remove the coils.
3 Topo-isomerase rejoins DNA strand.
DNA supercoil

(c)

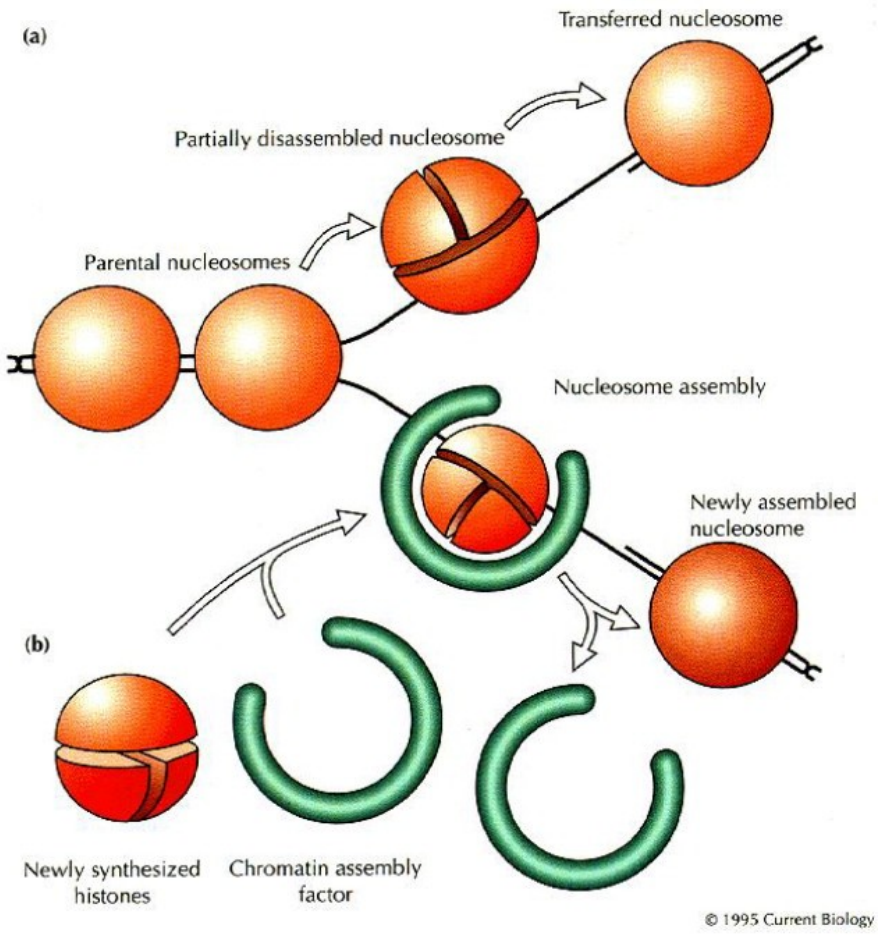
Chromatin: Nucleosome assembly during DNA replication

- DNA Packaging with Nucleosomes in Eukaryotic cells
- This particle comprises 146 bp of DNA wrapped in nearly two turns around an octamer of histone proteins



Chromatin Remodeling Proteins

- After a stretch of DNA is replicated, nucleosomes are reassembled on the newly formed strands
- *Nucleosome assembly protein-1* (*Nap-1*) and *chromatin assembly factor-1* (*CAF-1*) are examples of chromatin assembly proteins
- The dynamic disassembly/reassembly allows nucleosome association with DNA throughout the replication process

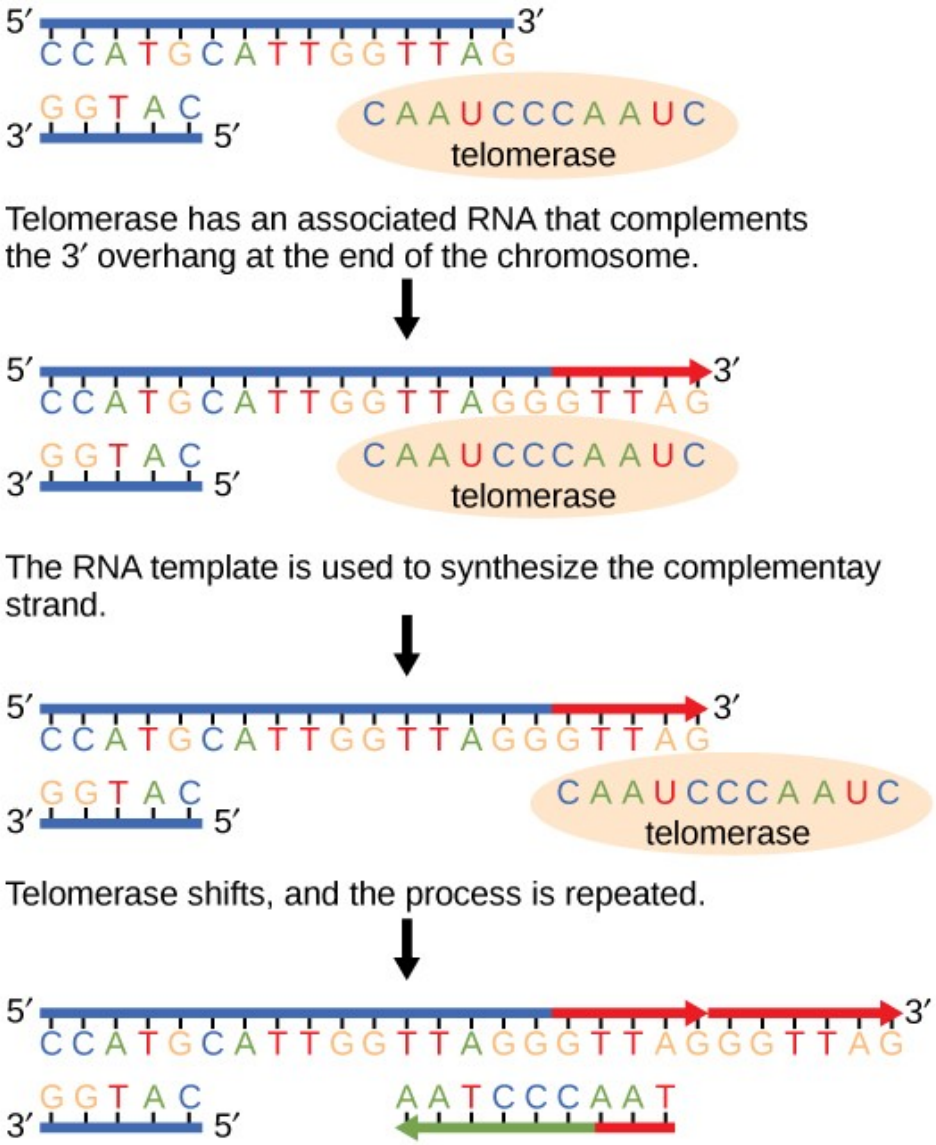


Telomeres Solve the DNA End-Replication Problem

- Linear DNA molecules have a problem in completing DNA replication
- Each round of replication would end with the loss of some nucleotides from the ends of each linear molecule
- Eukaryotes solve this problems with **telomeres,** highly repeated sequences at the ends of chromosomes

Telomeres and Telomerase

- Human telomeres have 100 to 1500 copies of TTAGGG at the ends of chromosomes
- These noncoding sequences ensure that the cell will not lose important genetic information if DNA molecules shorten during replication
- A polymerase called **telomerase** can catalyze the addition of repeats to chromosome ends
- https://www.biointeractive.org/classroom-resources/structure-and-function-telomeres


5′
3′
C C A T G C A T T G G T T A G
G G T A C
3′
5′
C A A U C C C A A U C
telomerase
Telomerase has an associated RNA that complements the 3′ overhang at the end of the chromosome.
5′
3′
C C A T G C A T T G G T T A G G G T T A G
G G T A C
3′
5′
C A A U C C C A A U C
telomerase
The RNA template is used to synthesize the complementay strand.
5′
3′
C C A T G C A T T G G T T A G G G T T A G
G G T A C
3′
5′
C A A U C C C A A U C
telomerase
Telomerase shifts, and the process is repeated.
5′
3′
C C A T G C A T T G G T T A G G G T T A G G G T T A G
G G T A C
3′
5′
A A T C C C A A T
Primase and DNA polymerase syntesize the complementary strand.

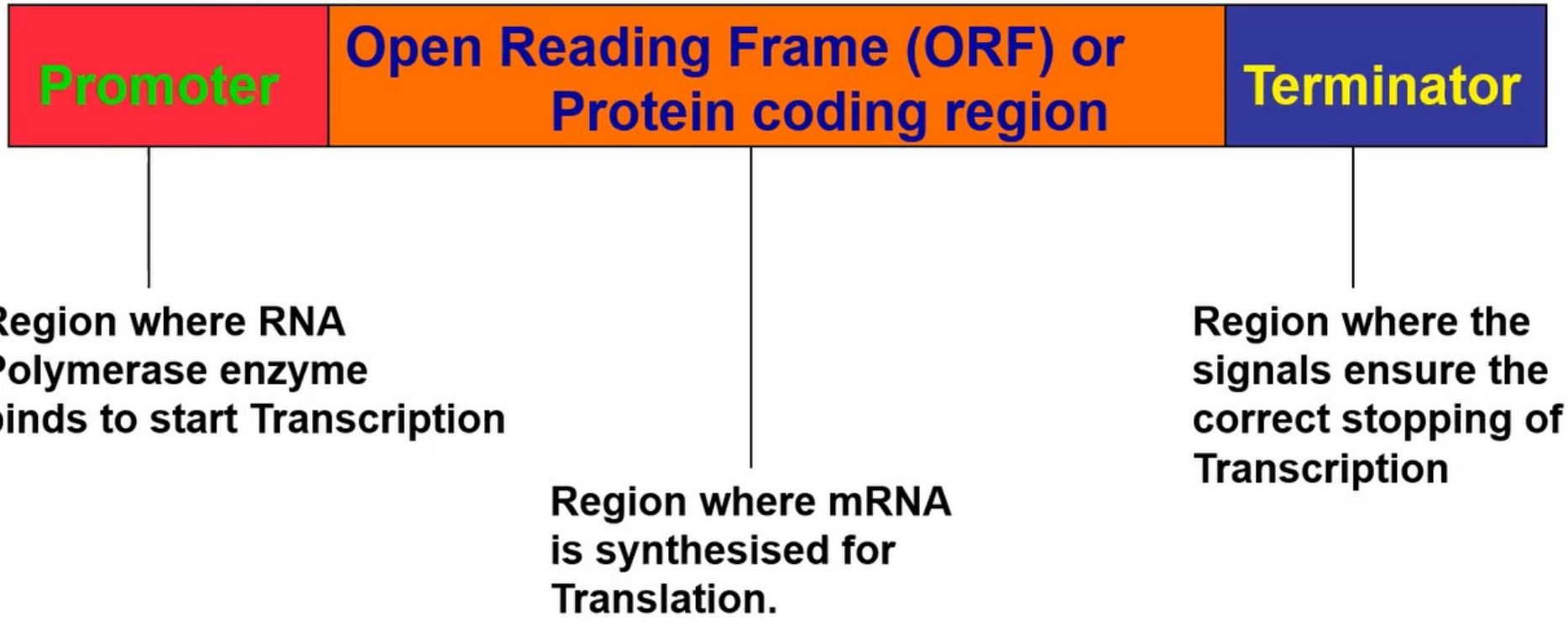
Rates of Replication

- **S phase** is very rapid in embryonic cells, where cell divisions occur in quick succession, but slower in adult cells, in which fewer and more widely spaced replicons are used
- During S phase in eukaryotes, **not all replicons are activated at the same time**; some are replicated early and others later
- **Genes that are transcriptionally active are replicated earlier than inactive genes.**

Errors during DNA Replication

DNA Damage and

Typical Eukaryotic Gene Structure



DNA Damage and Repair

- DNA must be accurately passed on to daughter cells
- In addition to ensuring that replication is faithful, this also means that DNA alterations must be repaired
- DNA alterations, or **mutations,** can arise spontaneously or through exposure to environmental agents

Mutations Can Occur Spontaneously During Replication

- During DNA replication, some types of mutations occur through
  - Spontaneous mispairing of bases
  - Slippage during replication (e.g. repeated regions)
  - Spontaneous damage to individual bases

Mutations and Translation

- mRNAs may contain mutant codons that cause errors in the polypeptide chain synthesized
- There are several types of mutations in which the DNA change involves only one or a few base pairs

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Other Types of Mutations

- Sometimes a single amino acid change may not affect a protein's function in a major way
- Substituting one amino acid for another may not affect the protein function if both amino acids are similar
- Some mutations may be *silent mutations* that affect the third base of the codon and results in no change of amino acid

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Indels

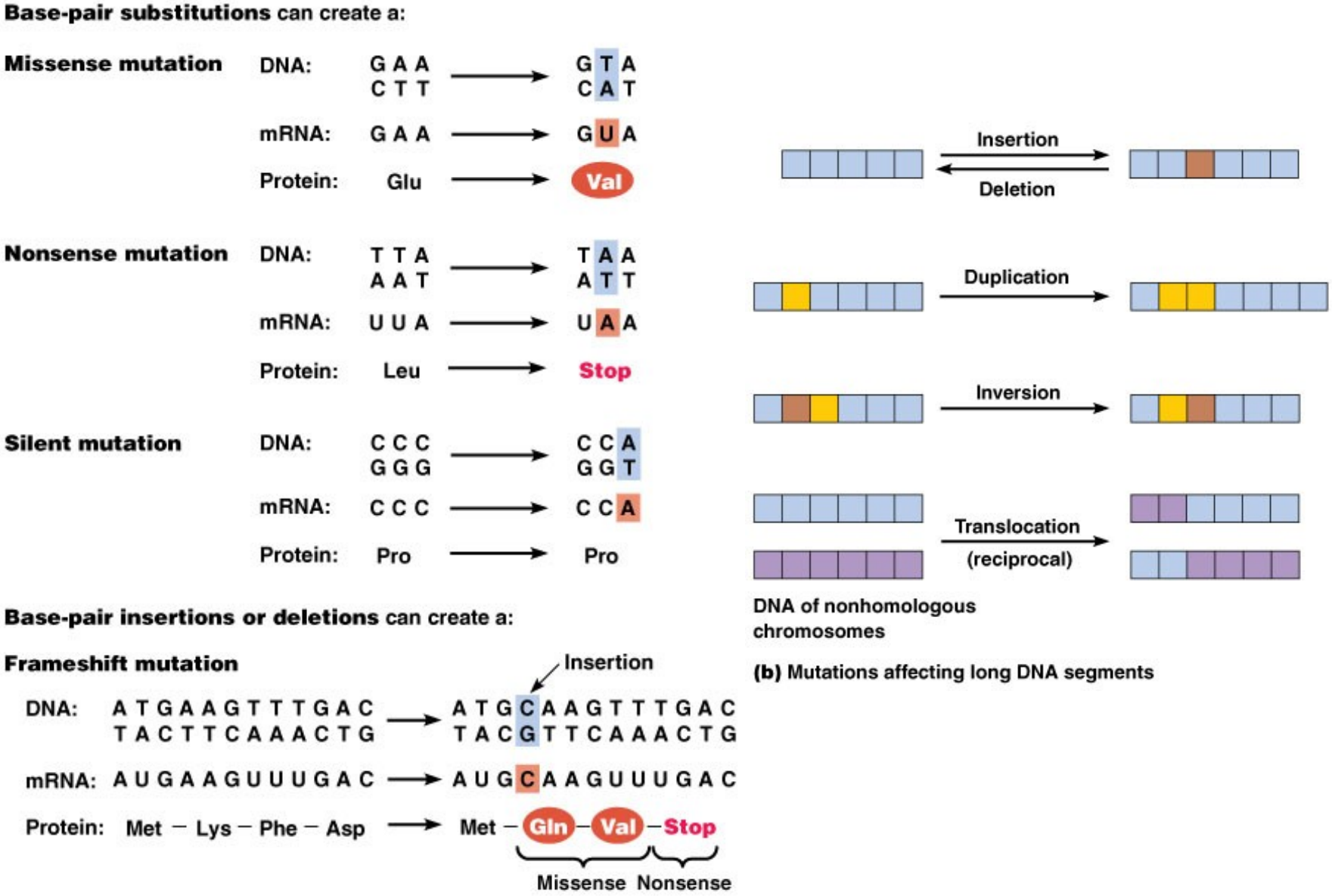
- Nonsense, nonstop, and missense codons can also arise from *base-pair insertions, deletions,* or a combination of these
- These are called *indels*, and they cause *frameshift mutations*

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Frameshift Mutations

- The gene is written in a language of three-letter words (triplet code)
- Inserting or deleting a nucleotide (**indel** mutations) causes the rest of the sequence to be read out of phase—this is a shift in the *reading frame*
- Mutations that cause insertion or deletion of a nucleotide are thus called **frameshift mutations**

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Mutagens Can Induce Mutations

- DNA damage can be caused by mutation-causing agents, **mutagens**
- Environmental mutagens fall into two categories: *chemicals and radiation*
- Mutation can also be induced by mobile genetic elements, such as found in viruses, or *transposable elements (transposons)*
- Mutagenic chemicals alter DNA structure through a variety of mechanisms

DNA Damage by Chemical Mutagens

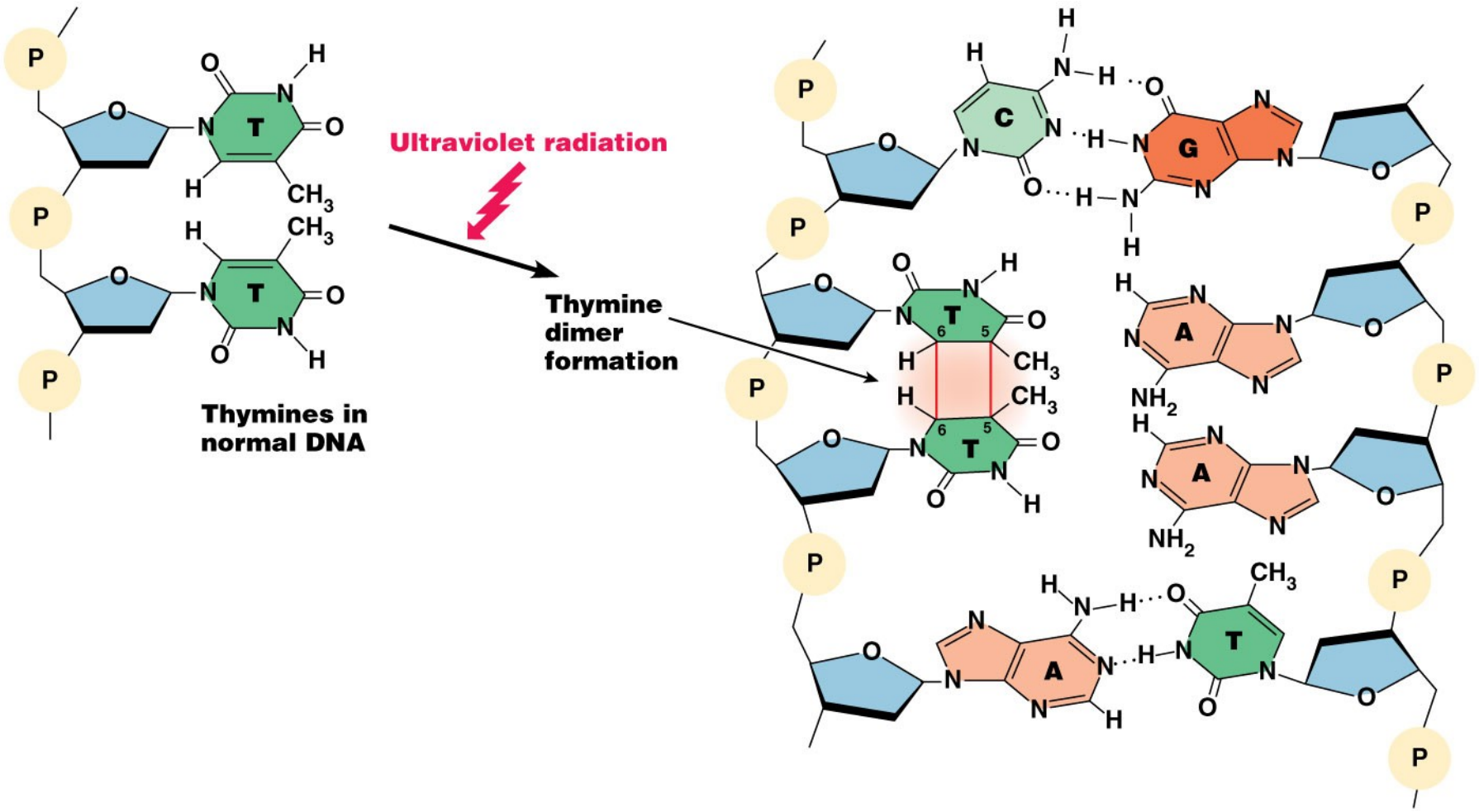
- *Base analogues* resemble nitrogenous bases and are incorporated into DNA
- *Base-modifying agents* react chemically with DNA bases to alter their structures, forming DNA adducts
- *Intercalating agents* insert themselves between adjacent bases, distorting DNA structure

- They all prevent interaction DNA with proteins like DNA Polymerase etc.

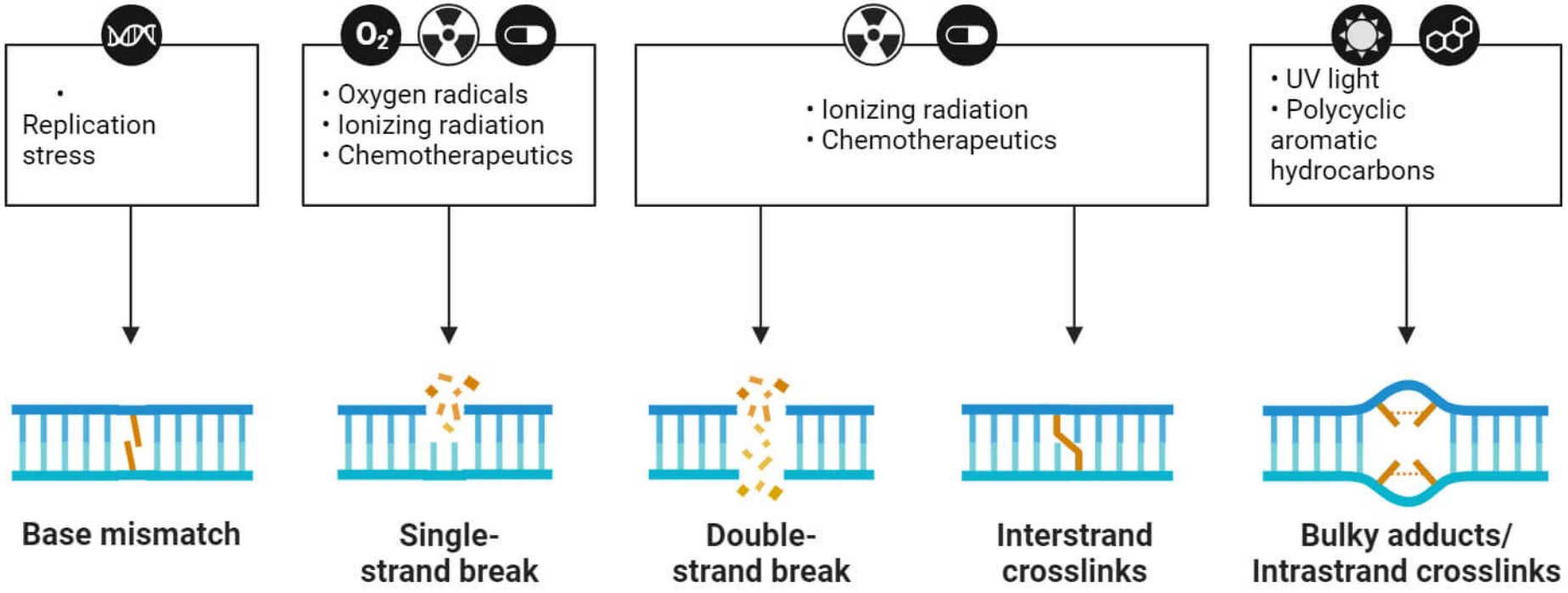
Radiation

- Ultraviolet radiation alters DNA by triggering pyrimidine dimer formation—*covalent bonds between adjacent pyrimidine bases (Thymidine dimers)*

- X-rays and related types of radiation, called ionizing radiation, remove electrons from molecules and generate highly reactive intermediates that damage DNA


Ultraviolet radiation
Thymine dimer formation
Thymines in normal DNA

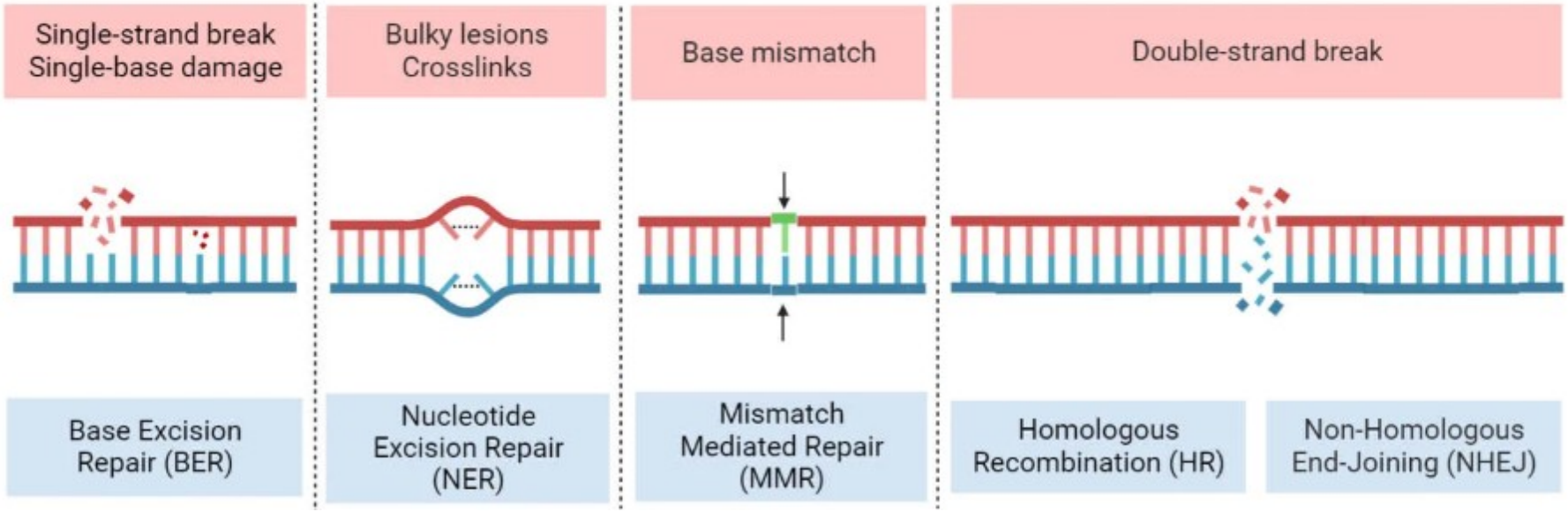
Common Causes of DNA Damage



DNA Repair Systems Correct Many Kinds of DNA Damage

- A variety of mechanisms have evolved for DNA repair
- The strategies depend on the damage type; how severe the damage is and whether or not the cell is undergoing division
  - DNA damage repair is carried out throughout a cell's life but is accelerated before DNA replication.

DNA Repair Mechanisms



Nucleotide Excision Repair

